

XRPL Validator — Daily Engineering Report • 2026-09-21

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **14 commits** (xrpl-validator 14)
- **1 engine finding fixed** (F356)
- 1 test / vector commit
- Amendment **PermissionDelegationV1_1** gained majority at ledger 107,135,489 (2026-09-21 11:18 UTC)
- Amendment **BatchV1_1** held majority (since 2026-09-15 14:06 UTC); activation due ~2026-09-29 14:06 UTC
- Amendment **PermissionDelegationV1_1** held majority (since 2026-09-21 11:18 UTC); activation due ~2026-10-05 11:18 UTC

Engine findings fixed

- **fix(ledger): F356 — EscrowFinish honours the destination's DepositAuth; campaign 14 vectors** (c3904b9 , xrpl-validator) Finding 356 (campaign 14, testnet CFD60CB8BA7A): rippled's EscrowFinish calls verifyDepositPreauth right after the tecNO_DST test (EscrowFinish.cpp:316) — a destination under lsfdDepositAuth accepts the finish only from itself, from an account it preauthorized, or through credentials naming a DepositPreauth object; otherwise tecNO_PERMISSION. Our EscrowFinish never asked and paid the escrow out.

Engineering changes

- **fixCleanup3_4_0: port the amendment's transactor-side rules behind the ledger's Amendments gate** (c6b4fd3 , xrpl-validator) Ported (each gated on amendments::fix_cleanup_3_4_0, unit-tested; no oracle exists — the vendored libxrpl predates the amendment and testnet has it off): - EscrowFinish/EscrowCancel: the escrow's reserve is recycled before the delivery/return re-creates a line (EscrowFinish.cpp:345, EscrowCancel.cpp:174); EscrowCancel also gains the pre-existing createAsset reserve check. - OfferCreate + Payment DomainID: ...

Tests and campaigns

- **test(node): campaign 13 — 15 testnet NFT depth vectors + NFT read-set hydration** (21de443 , xrpl-validator)

Docs and operations

- **Findings 348-349:** CheckCash judges the cashier's line (auth, freeze) after the funds test; credentials::valid runs in EscrowFinish, PaymentChannelClaim, AccountDelete (7fbf877 , xrpl-validator)
- **Finding 350:** a pseudo-account is created with Sequence 0 once SingleAssetVault or LendingProtocol is enabled (devnet #5488138); mainnet keeps the ledger sequence until then (d74d5b3 , xrpl-validator)
- **amm:** Finding 350 marker capitalised for the deploy marker grep (9428cef , xrpl-validator)
- **Batch:** seven campaign-8 devnet vectors (all four modes, failures at every position, all-revert, multi-account BatchSigners, offer create+cancel); an AllOrNothing revert files no inner. **Finding 351:** Clawback's preclaim order (a1cdddf , xrpl-validator)
- **Finding 352:** AMMClawback ported from rippled's applyGuts and equalWithdrawMatchingOneAmount; harness hydrates the holder's asset lines (79158d3 , xrpl-validator)
- **harness:** the probe hydrates the holder's line for Clawback (its whole preclaim read-set) (5c204d8 , xrpl-validator)
- **Finding 353:** EscrowFinish verifies the PREIMAGE-SHA-256 crypto-condition — tecCRYPTOCONDITION_ERROR for a wrong fulfillment, a missing or different Condition, or a Condition on an unconditioned escrow (faf89c6 , xrpl-validator)
- **harness:** the probe hydrates the AMMBid bidder's LP line and the AuthAccounts roots (a05696d , xrpl-validator)
- **harness:** the probe hydrates the AMMBid bidder's LP line and the AuthAccounts roots (f48d696 , xrpl-validator)
- **Finding 354:** AccountDelete removes DID, Oracle, Delegate and Credential objects (rippled's nonObligationDeleter); harness hydrates the whole owner directory for AccountDelete (d4b5158 , xrpl-validator)
- **Finding 355:** AccountSet cannot clear GlobalFreeze under NoFreeze, and NoFreeze has no clear at all — both silent no-ops (SetAccount.cpp:346-375); nine campaign-12 TrustSet/AccountSet vectors (0787019 , xrpl-validator)

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Voting events:

- PermissionDelegationV1_1 gained majority — ledger 107,135,489, 2026-09-21 11:18 UTC

Amendment	Majority since	Activation due
BatchV1_1	2026-09-15 14:06 UTC	2026-09-29 14:06 UTC
PermissionDelegationV1_1	2026-09-21 11:18 UTC	2026-10-05 11:18 UTC

Amendments enabled on mainnet: 94.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **AMM Self-Referential Volatility Fee Bounds** (amm-defi) Add sfFeeMin/sfFeeMax bounds and a bounded 8-entry sfPriceHistory circular buffer (STArray of {sfLedgerSeq, sfPrice}) to the AMM ledger object, settab
- **On-Ledger Finality Certificate Pseudo-Transaction** (consensus) Add a new pseudo-transaction ttFINALITY_CERT, injected by validators at ledger close in the same class as the existing NegativeUNL ttUNL_MODIFY, that
- **Spend-Policy Enforcement on PayChan Claims** (custody) Extend the PayChanClaim transactor's preclaim/doApply to consult the claimant's AccountPolicy object (g18's owner-reserved ltACCOUNT_POLICY, keylet::a

Appendix — commits

Time	Repo	Commit	Subject
04:56	xrpl-validator	c6b4fd3	fixCleanup3_4_0: port the amendment's transactor-side rules behind the ledger's Amendments gate
07:56	xrpl-validator	7fbf877	Findings 348-349: CheckCash judges the casher's line (auth, freeze) after the funds test; credentials::valid runs in EscrowFinish, PaymentChannelClaim, AccountDelete
08:08	xrpl-validator	d74d5b3	Finding 350: a pseudo-account is created with Sequence 0 once SingleAssetVault or LendingProtocol is enabled (devnet #5488138); mainnet keeps the ledger sequence until then
08:09	xrpl-validator	9428cef	amm: Finding 350 marker capitalised for the deploy marker grep
09:05	xrpl-validator	a1cdddf	Batch: seven campaign-8 devnet vectors (all four modes, failures at every position, all-revert, multi-account BatchSigners, offer create+cancel); an AllOrNothing revert files no inner. Finding 351: Clawback's preclaim order
09:50	xrpl-validator	79158d3	Finding 352: AMMClawback ported from rippled's applyGuts and equalWithdrawMatchingOneAmount; harness hydrates the holder's asset lines
09:54	xrpl-validator	5c204d8	harness: the probe hydrates the holder's line for Clawback (its whole preclaim read-set)
10:16	xrpl-validator	faf89c6	Finding 353: EscrowFinish verifies the PREIMAGE-SHA-256 crypto-condition — tecCRYPTOCONDITION_ERROR for a wrong fulfillment, a missing or different Condition, or a Condition on an unconditioned escrow
10:19	xrpl-validator	a05696d	harness: the probe hydrates the AMMBid bidder's LP line and the AuthAccounts roots
10:20	xrpl-validator	f48d696	harness: the probe hydrates the AMMBid bidder's LP line and the AuthAccounts roots
10:53	xrpl-validator	d4b5158	Finding 354: AccountDelete removes DID, Oracle, Delegate and Credential objects (rippled's nonObligationDeleter); harness hydrates the whole owner directory for AccountDelete
11:27	xrpl-validator	0787019	Finding 355: AccountSet cannot clear GlobalFreeze under NoFreeze, and NoFreeze has no clear at all — both silent no-ops (SetAccount.cpp:346-375); nine campaign-12 TrustSet/AccountSet vectors
11:50	xrpl-validator	21de443	test(node): campaign 13 — 15 testnet NFT depth vectors + NFT read-set hydration
12:43	xrpl-validator	c3904b9	fix(ledger): F356 — EscrowFinish honours the destination's DepositAuth; campaign 14 vectors